



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-8

Depth First Search (DFS)

Aim

To traverse a graph using the DFS algorithm.

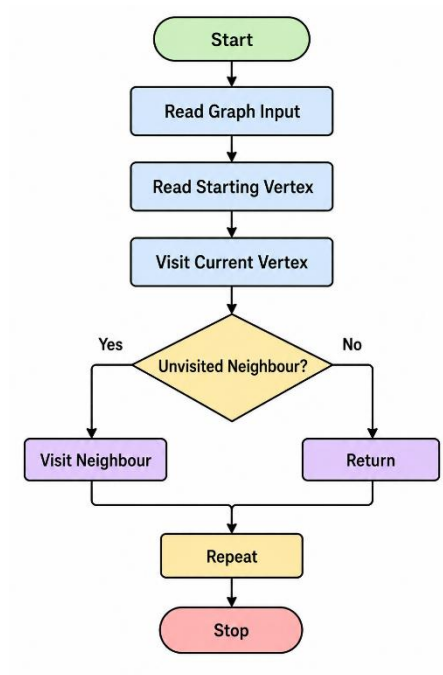
Objective

To traverse a graph using the DFS algorithm.

Algorithm

1. Start.
2. Read the number of vertices and edges.
3. Create the graph using an adjacency list.
4. Read the starting vertex.
5. Mark the starting vertex as visited.
6. Visit the current vertex and recursively visit all its unvisited adjacent vertices.
7. Repeat until all reachable vertices are visited.
8. Stop.

Flowchart



Code

```
graph = {}
```

```
n = int(input("Enter number of vertices: "))
```

```
for _ in range(n):
```

```
    v = input("Enter vertex: ")
```

```
    graph[v] = input(f"Enter neighbours of  
{v} (space-separated): ").split()
```

```
start = input("Enter starting vertex: ")
```

```
visited = set()
```

```
def dfs(node):
```

```
    if node not in visited:
```

```

print(node, end=" ")

visited.add(node)

for neighbour in graph[node]:

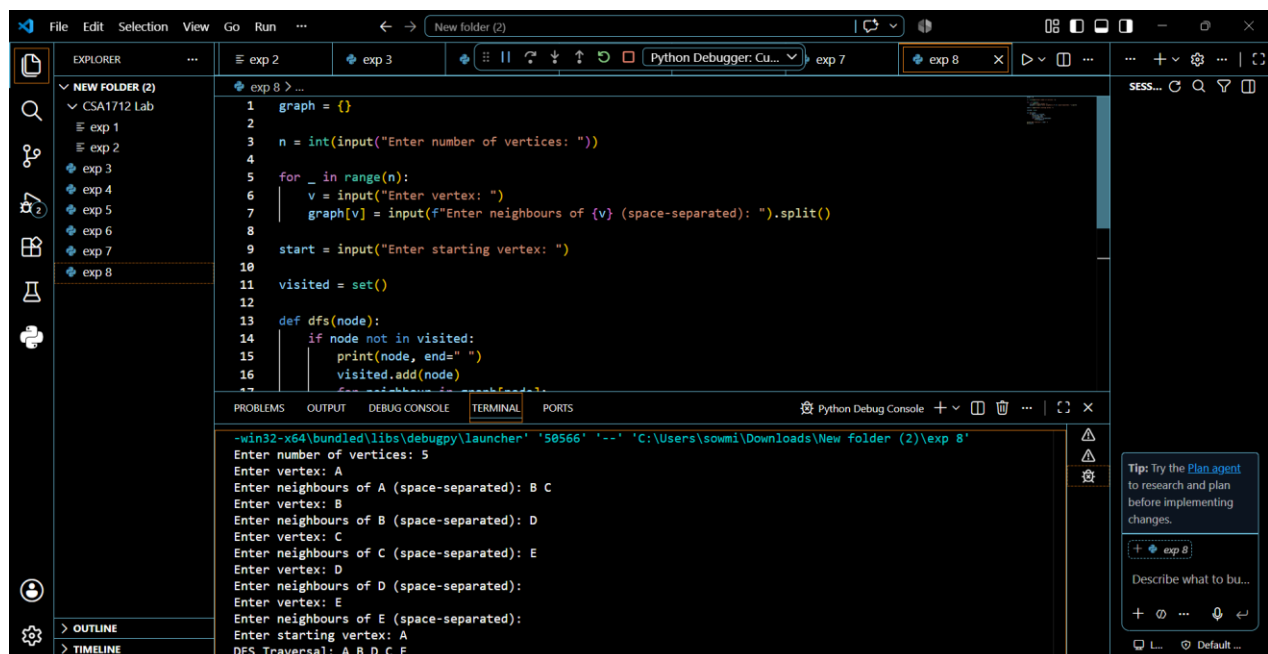
    dfs(neighbour)

print("DFS Traversal:", end=" ")

dfs(start)

```

Output



The screenshot shows a Python IDE with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The code editor displays the DFS algorithm implementation. The terminal shows the execution of the program, including user input for the number of vertices, the graph structure, the starting vertex, and the resulting DFS traversal output.

```

1 graph = {}
2
3 n = int(input("Enter number of vertices: "))
4
5 for _ in range(n):
6     v = input("Enter vertex: ")
7     graph[v] = input(f"Enter neighbours of {v} (space-separated): ").split()
8
9 start = input("Enter starting vertex: ")
10
11 visited = set()
12
13 def dfs(node):
14     if node not in visited:
15         print(node, end=" ")
16         visited.add(node)
17         for neighbour in graph[node]:
18             dfs(neighbour)
19
20 dfs(start)

```

```

- win32-x64\bundle\libs\debugpy\launcher '50566' '--' 'C:\Users\sowmi\Downloads\New folder (2)\exp 8'
Enter number of vertices: 5
Enter vertex: A
Enter neighbours of A (space-separated): B C
Enter vertex: B
Enter neighbours of B (space-separated): D
Enter vertex: C
Enter neighbours of C (space-separated): E
Enter vertex: D
Enter neighbours of D (space-separated):
Enter vertex: E
Enter neighbours of E (space-separated):
Enter starting vertex: A
DFS Traversal: A B D C E

```

Result

The graph was traversed successfully using the DFS algorithm.

Conclusion

The DFS algorithm successfully visits all reachable vertices by exploring each branch completely before backtracking.